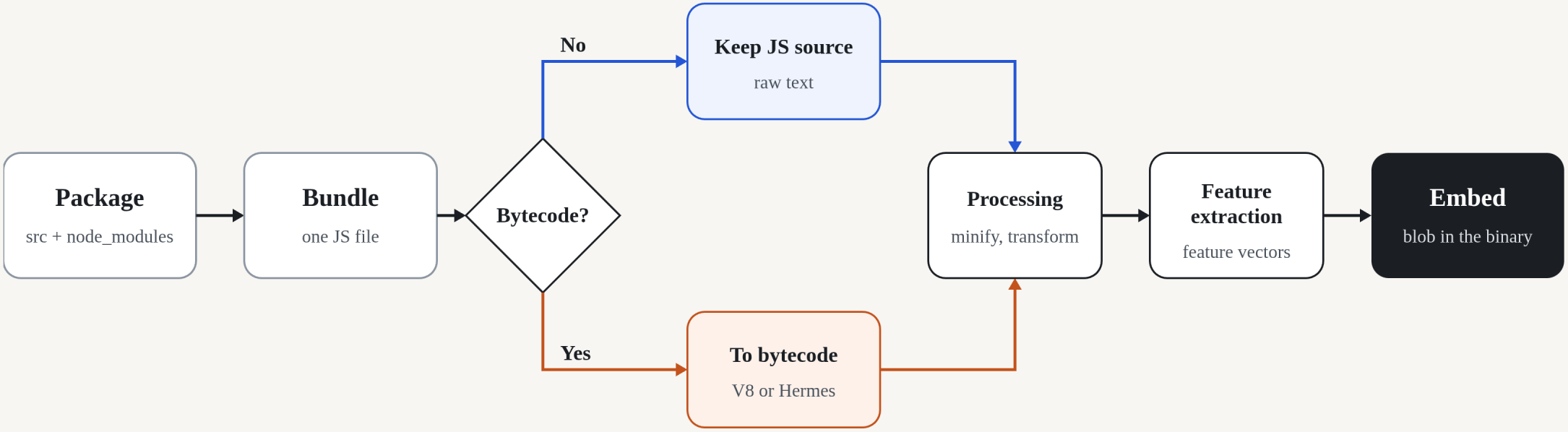
JavaScript packaging

From package to binary

Bundle the code, optionally compile it to bytecode, then embed it.

Pipeline graph · Tradeoffs

One decision before embedding



Blue: raw JS source. Orange: bytecode. Both meet at Processing before embedding.

If the bytecode path is chosen

Many bundles, each functional

Bundle automatically

bundle a large number of packages, and discard what failed.

Each bundle is functional

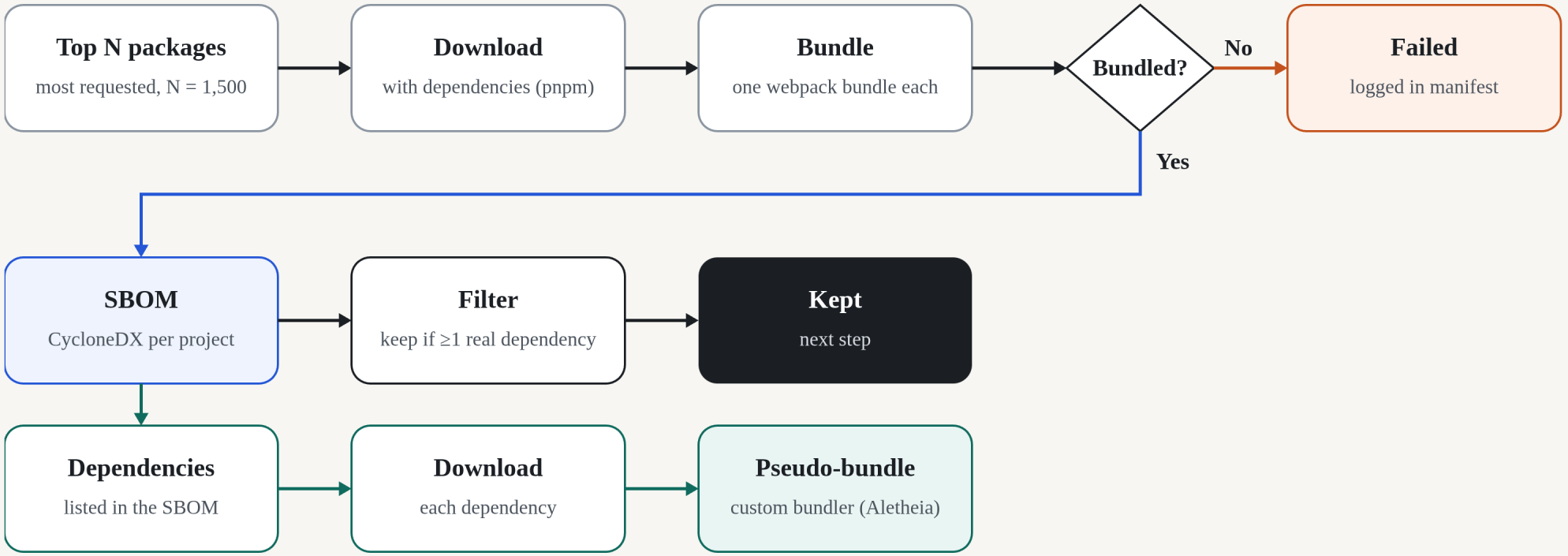
Every bundle must be valid js so it can be compiled to bytecode.

A custom bundler does not fit

Aletheia Bundler does not fit this solution

Compiled bundles then flow into Processing, feature extraction, and Embed.

From top packages to a filtered set



Blue: bundled path. Teal: SBOM dependencies, re-bundled with the custom bundler. Orange: failed, logged for review. Only about 1,200 of the packages bundled.

Questions

1. Why a custom bundler instead of Webpack?
2. When a package gets a new version, does it mostly add code, change existing code, or both?
3. How should I configure the bundling, and what should guide the choice?